PHYLOGENETIC DISTANCE COMPUTATION USING EULER TOUR TREES

Introduction

- This presentation explores the application of Euler Tour Trees (ETTs) in bioinformatics for efficient

phylogenetic distance computation, with a focus on their role in tools like Nextstrain for

real-time pathogen tracking.

- ETTs, a pivotal data structure in dynamic graph algorithms,

enable rapid queries such as lowest common ancestor (LCA) and path distances in evolutionary trees.

Introduction to Euler Tour Trees

- Euler tour tree, or ETT, is a smart dynamic tree that can be maintained with a brute-force data structure.
- Euler tour tree (ETT) technique is a method for representing a rooted undirected tree as a number sequence
- There are several common ways to build this representation.
- the first is called the Euler tour, which has been implemented for this project.
- **Euler tour** is defined as a way of traversing tree such that each vertex is added to the tour when we visit it (either moving down from parent vertex or returning from child vertex).
- We start from root and reach back to root after visiting all vertices.

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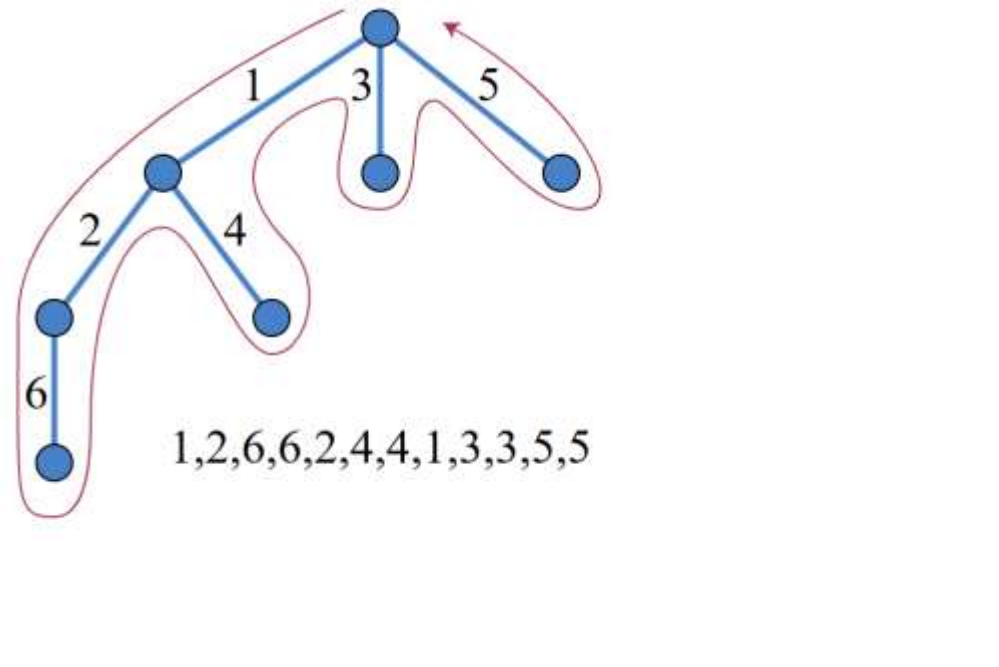
Content

- Euler Tour Trees represent a tree by linearizing its structure through a depth-first search (DFS), recording each node's entry and exit times.
- This creates a sequence of length 2n−1 for an n-node tree, where subtrees are contiguous subarrays [1].

Euler Tour Technique



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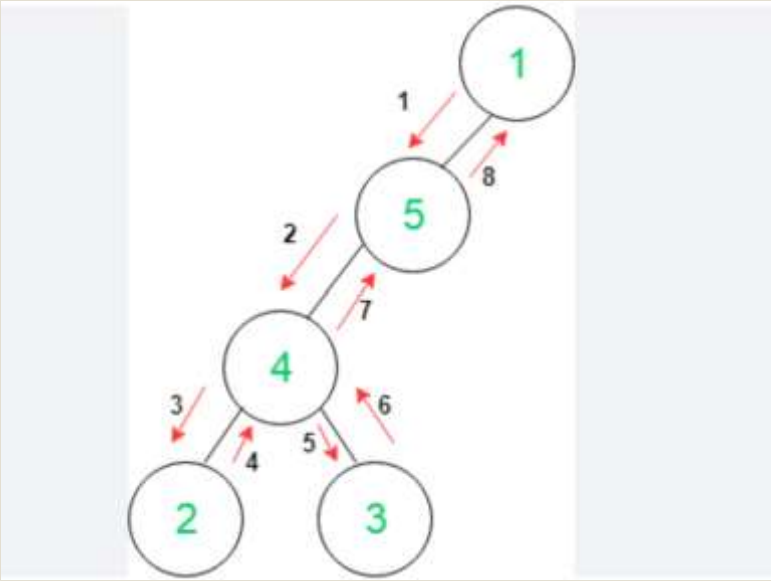
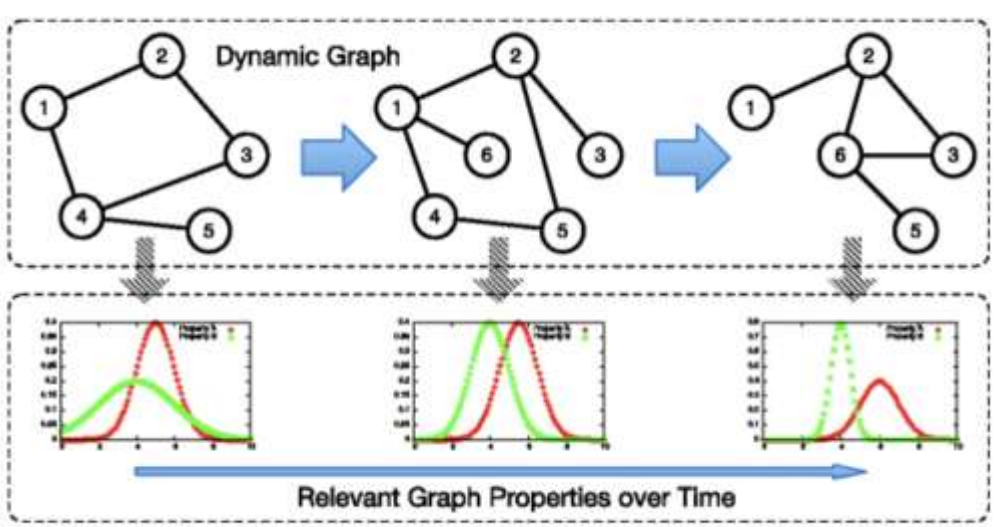
- Queries like subtree aggregates or LCAs reduce to range queries, often using sparse tables for O(1) efficiency after O(n log n) preprocessing.
- ETTs are foundational for dynamic trees, supporting operations like linking (adding edges) and cutting (removing edges) in amortized O(log n) time when paired with balanced binary search trees (e.g., splay trees) [2].
- This makes them critical for dynamic graphs, particularly forests, where connectivity evolves over time.



RELATION TO DYNAMIC GRAPHS

- Dynamic graphs undergo updates such as edge insertions or deletions while supporting queries like connectivity or distances.

- ETTs excel in dynamic trees, maintaining acyclicity and enabling efficient updates via structures like link-cut trees [2].

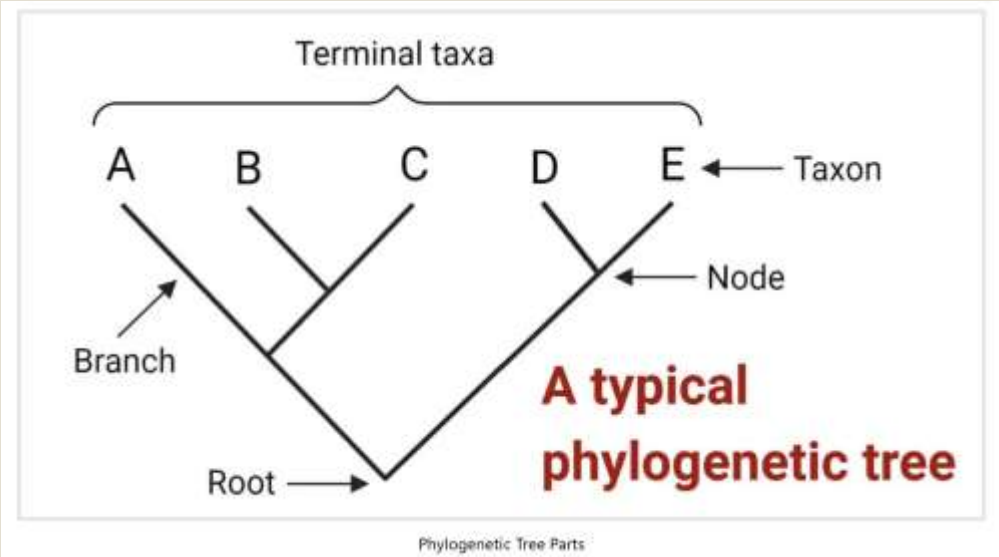


General application scenario of dynamic graph analysis

APPLICATION IN BIO-INFORMATICS:PHYLOGENETIC ANALYSIS

- Phylogenetic trees model evolutionary relationships, with nodes as taxa and edges as divergence, weighted by branch lengths (e.g., Mutations or time)

- ETTs reduce LCA to a range minimum query (RMQ) on the tour's depth array [1].



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- A key task is computing phylogenetic distances—the sum of branch lengths between nodes—via their LCA: distance(u, v) = depth[u] + depth[v] − 2 × depth[LCA(u, v)].

- Tools like Genesis and Gappa leverage ETTs for LCA and distance calculations in large phylogenies, supporting monophyly testing and split representations [4, 5].

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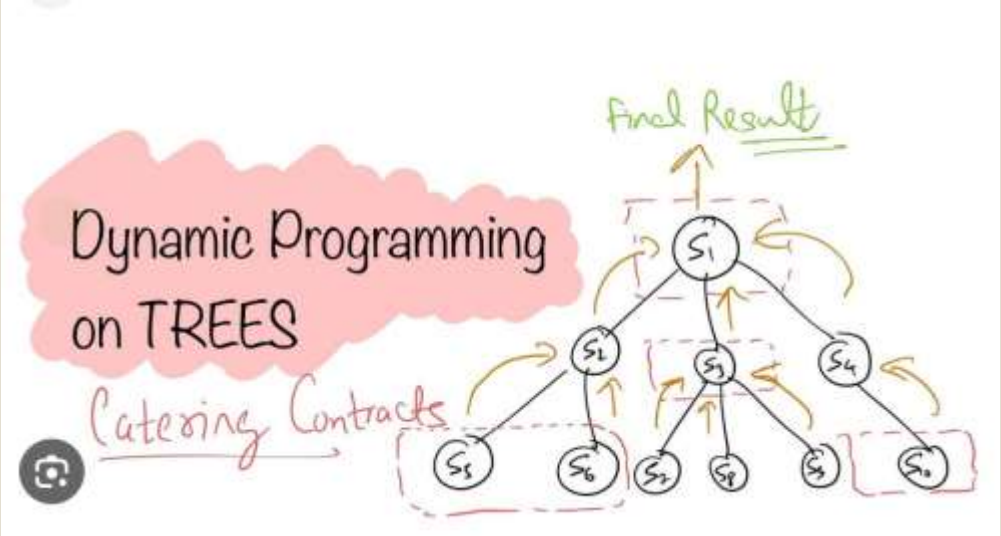
- These handle multifurcations and arbitrary node/edge data, crucial for complex trees. Dynamic updates, such as adding new sequences, align with ETTs' strengths in dynamic graphs, enabling efficient tree modifications [4].



SPECIFIC EXAMPLE-NEXTSTRAIN AND PATHOGEN TRACKING

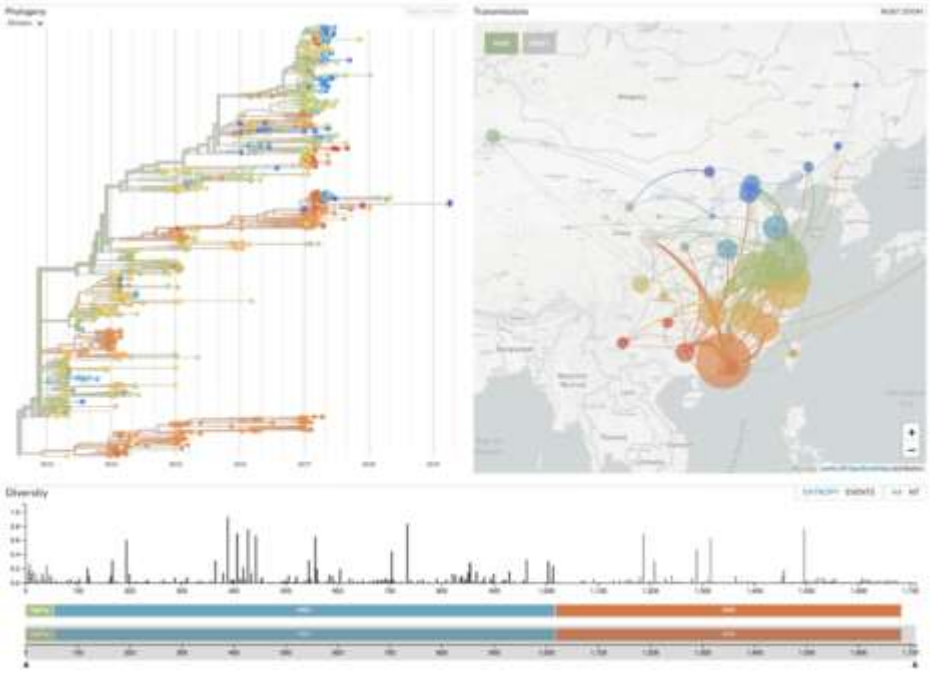
- Nextstrain, an open-source platform used by the WHO and CDC, tracks pathogen evolution in real time, such as for SARS-CoV-2 [6].

- It employs workflows like Augur for tree inference and Auspice for visualization, handling millions of sequences [7].

Nextstrain

Auspice

version v2.52.1



Auspice being used to show the spread of influenza H7N9 virus across Asia.

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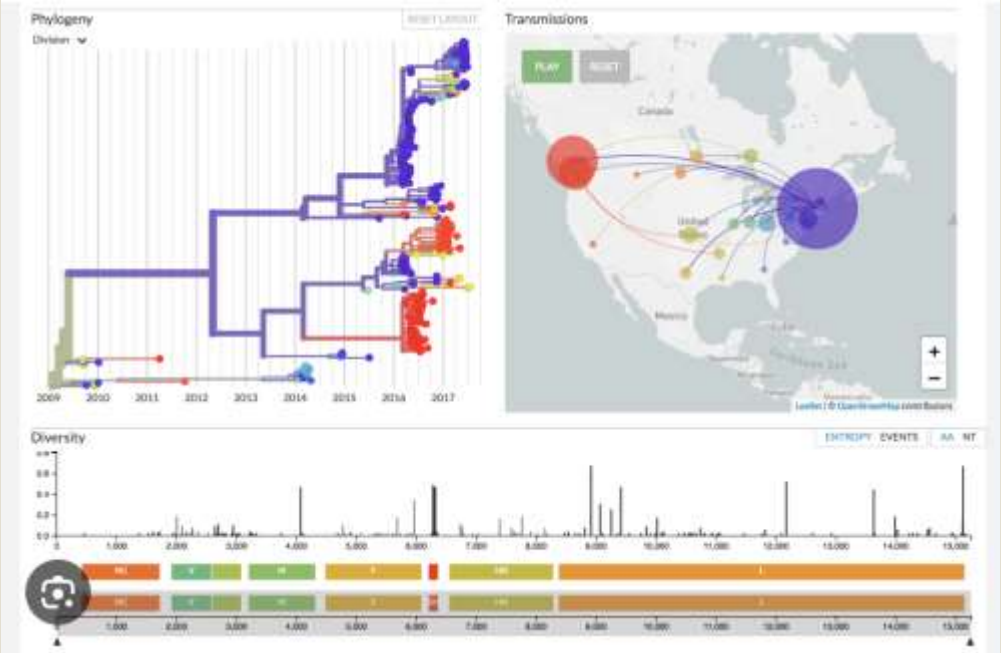
- Efficient queries are critical for interactive clade rendering and distance calculations.

- While Nextstrain uses JSON-based tree representations [8], tools like UShER integrate dynamic sample placement, adding sequences without rebuilding trees [9].



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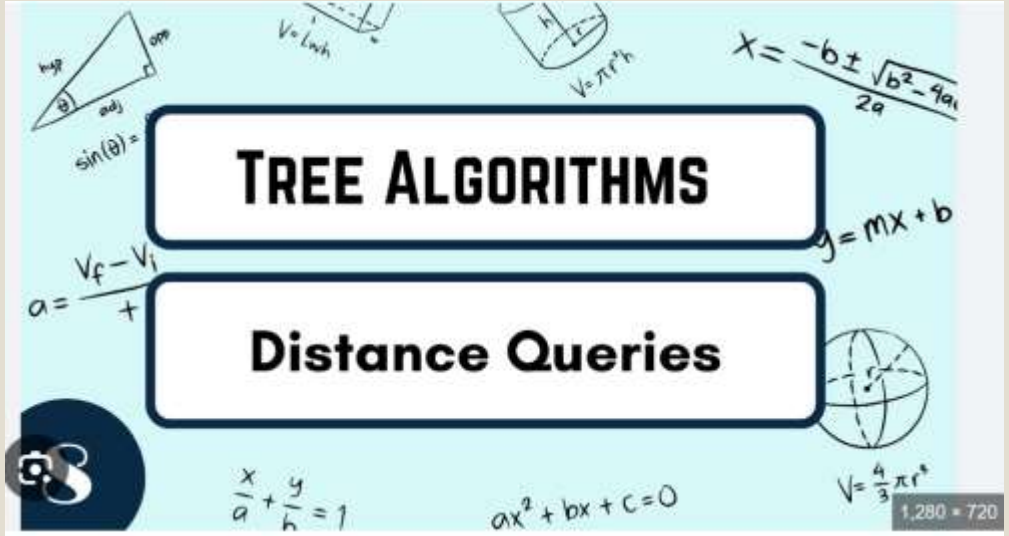
- ETT-like structures optimize LCA and distance queries in such dynamic settings, supporting cluster identification and mutation tracking during pandemics [8, 10].



OTHER APPLICATIONS OF ETT:

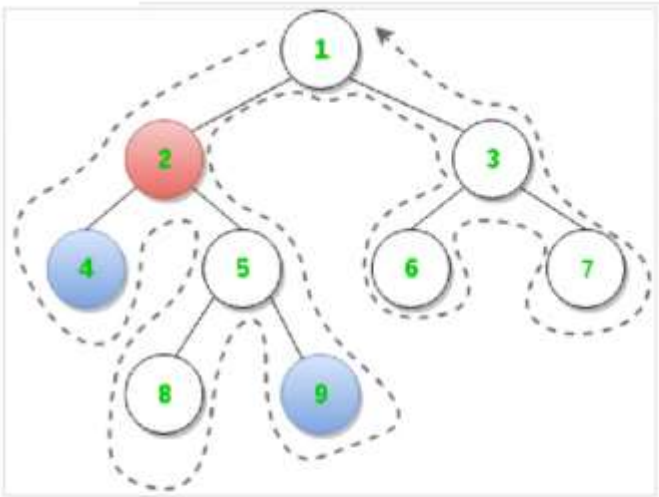
- Euler Tour Trees are used for :

❑ dynamic connectivity in networks

❑ efficiently answering subtree queries (e.g., subtree sums)

❑ finding Lowest Common Ancestors (LCA)

❑ updating nodes or paths within a changing tree structure



Euler Tour

An euler tour of the tree starting from node 1 will yield:

1	2	4	2	5	8	5	9	5	2	1	3	6	3	7	3	1

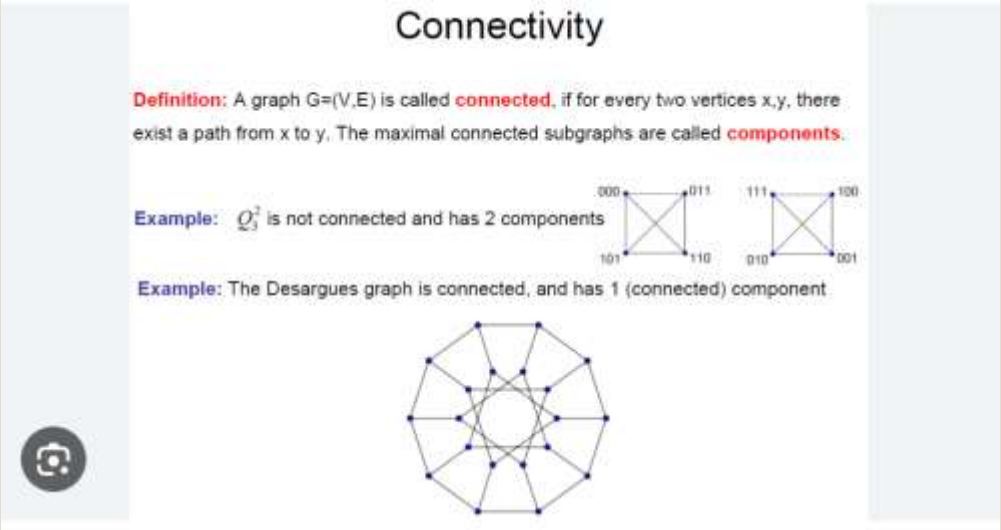
The corresponding levels for every node in Euler tour:

0	1	2	1	2	3	2	3	2	1	0	1	2	1	2	1	0

Dynamic Connectivity in networks

- In graph theory, dynamic connectivity refers to the problem of maintaining information about a graph's connected components and paths as edges are added or removed over time.

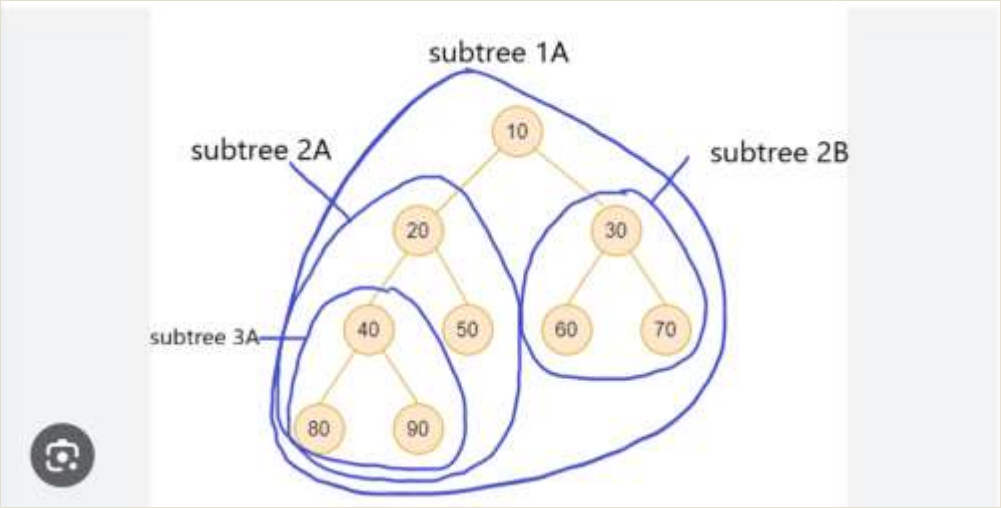
- Instead of analyzing a static snapshot, it involves developing data structures that efficiently support queries like "Are vertices X and Y connected?" after modifications to the graph's edges.



Subtree sum calculation

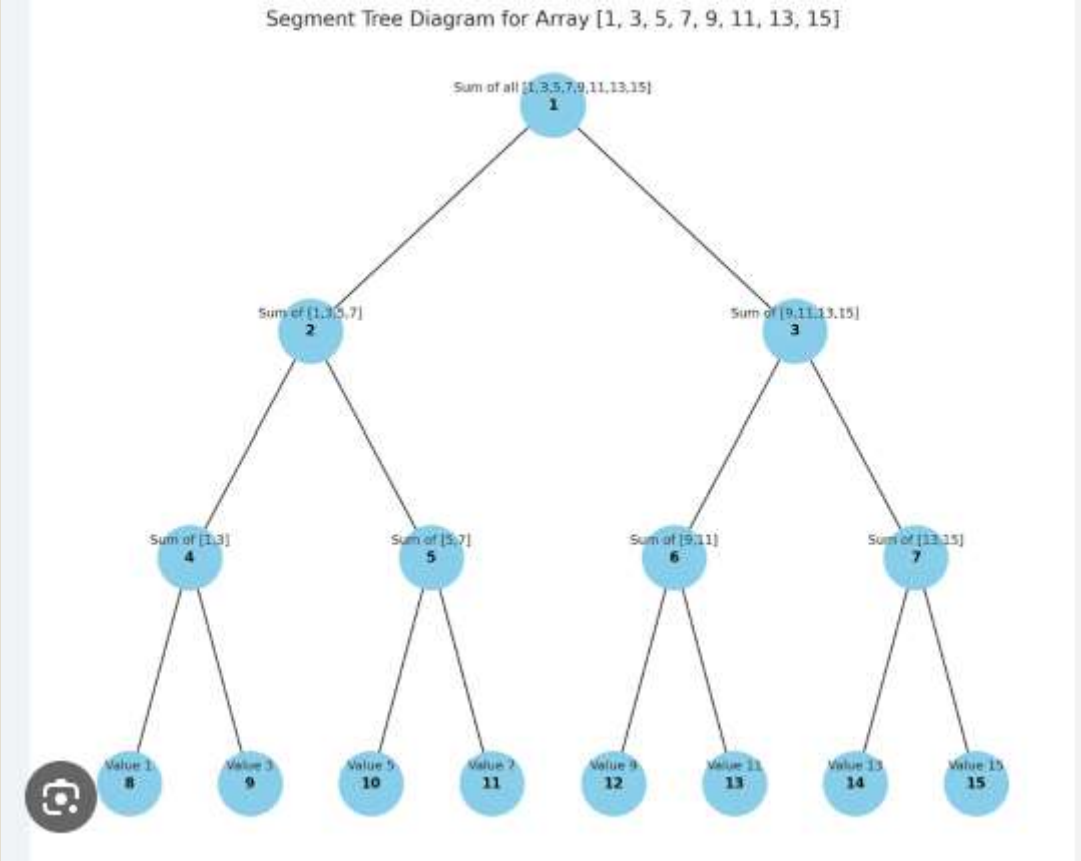
- Efficiently answering subtree queries using an Euler Tour involves transforming the tree structure into a linear array, allowing range queries on this array to correspond to subtree queries in the original tree.

- Benefits:
- Converts tree problems into efficient range queries on a linear data structure.
- Allows for logarithmic time complexity for many subtree queries and updates, especially when combined with Segment Trees or Fenwick Trees.



Updating Nodes

- Updating a node in an Euler tour tree typically involves finding the node's position in the flattened tour array (using its entry and exit times).
- Then performing a point update on the associated balanced binary search tree (like a BBST or segment tree) that represents the tour.
- Finally reflecting that change in the tree's aggregate data structure for queries.



CONCLUSION

- Euler Tour Trees bridge dynamic graphs and bioinformatics, offering efficient, updatable representations for phylogenetic queries. Their application in tools like Nextstrain enhances real-time epidemiology, informing global health responses.
- As genomic datasets grow, ETTs will remain pivotal for scalable analytics.

REFERENCES

- [1] Henzinger, M., King, V., "Improved deterministic algorithms for dynamic connectivity problems," Journal of Algorithms, 1999.
- [2] Sleator, D. D., Tarjan, R. E., "Self-adjusting binary search trees," Journal of the ACM, 1985.
- [3] Felsenstein, J., Inferring Phylogenies, Sinauer Associates, 2004.
- [4] Czech, L., et al., "Gappa: A library for phylogenetic placement analysis," Bioinformatics, 2022.
- [5] Czajkowski, M., et al., "Genesis: A toolkit for phylogenetic analysis," Bioinformatics, 2022.

CONTD ……

[6] Hadfield, J., et al., “Nextstrain: Real-time tracking of pathogen evolution,” Bioinformatics, 2018.

[7] Huddleston, J., et al., “Augur: A bioinformatics toolkit for phylogenetic analyses,” Bioinformatics, 2021.

[8] Nextstrain Team, “Nextstrain documentation,” https://docs.nextstrain.org, 2023.

[9] Turakhia, Y., et al., “Ultrafast sample placement on existing trees,” Bioinformatics, 2021.

[10] Hodcroft, E. B., et al., “Real-time tracking of SARS-CoV-2 evolution,” Nature, 2021.

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